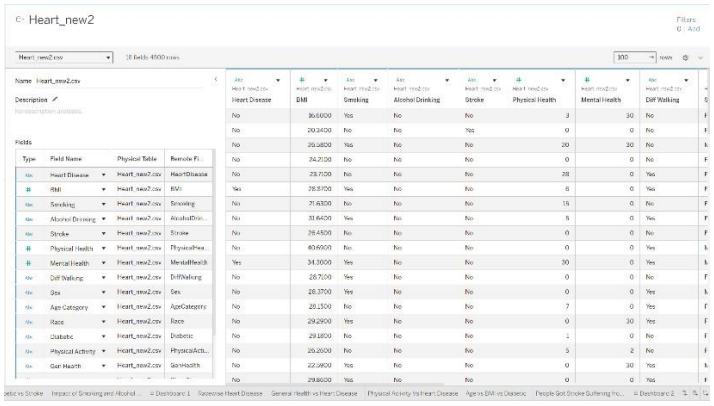
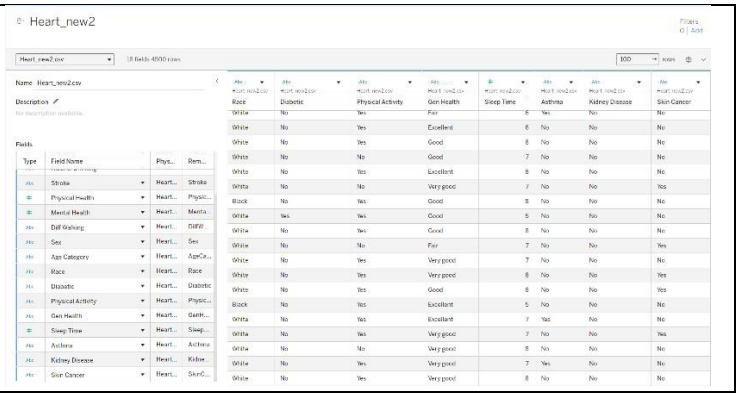
Project Development Phase
Performance Test

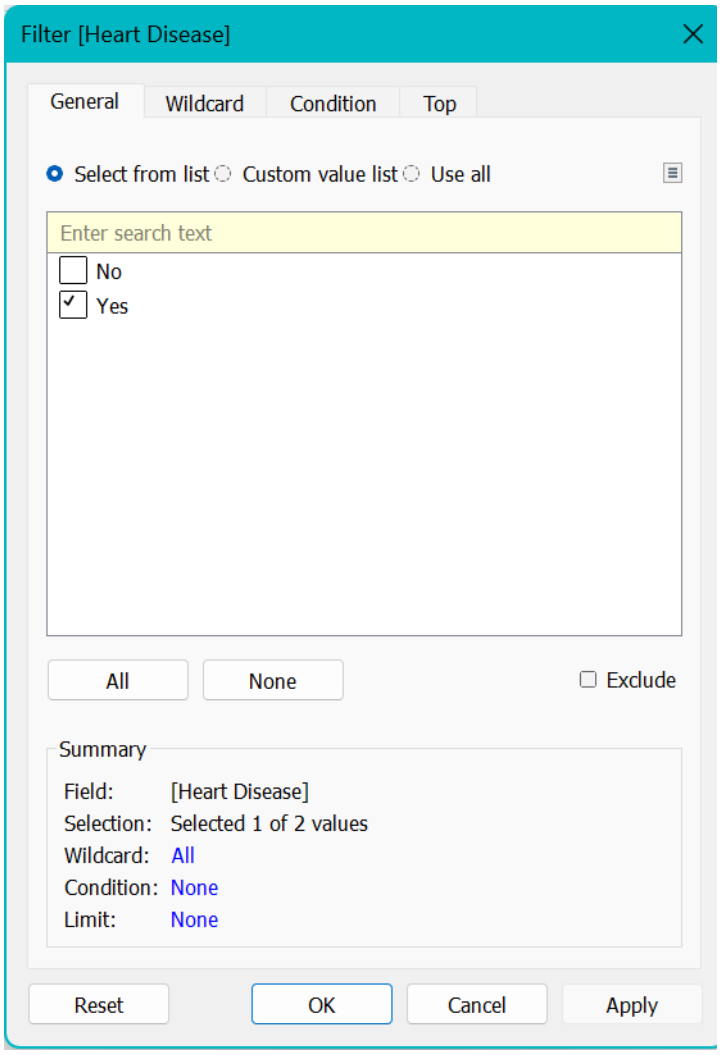
Date	07-03-2026
Team ID	Srashti Patel
Project Name	Heart Disease Analysis
Maximum Marks	5

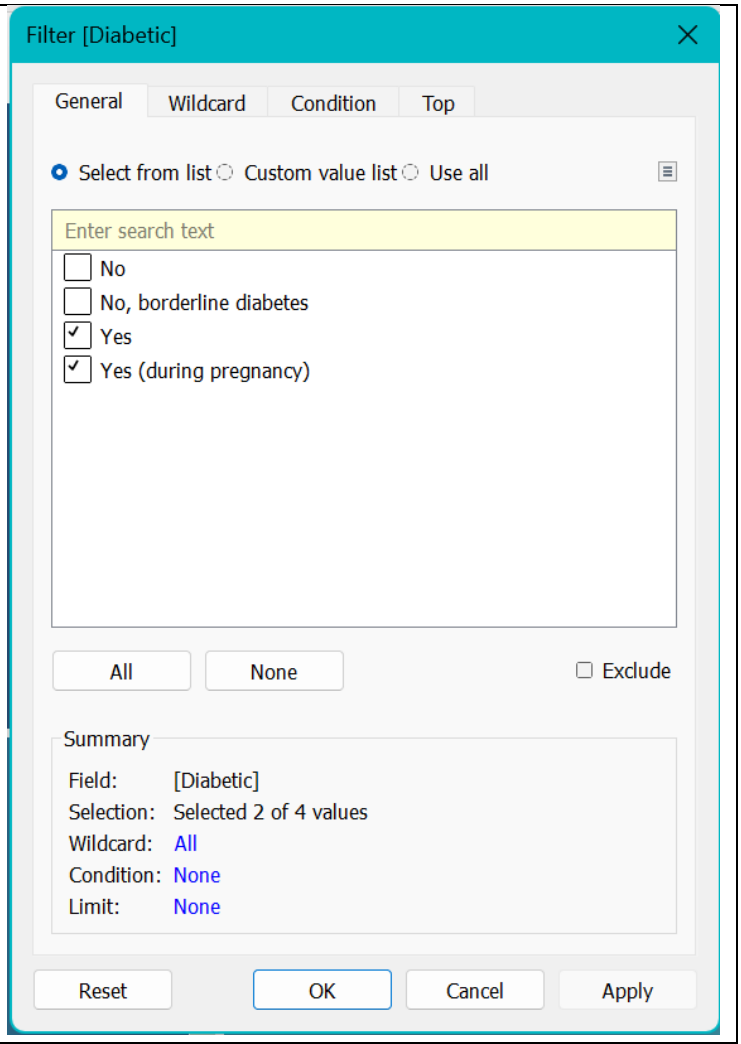
Model Performance Testing:

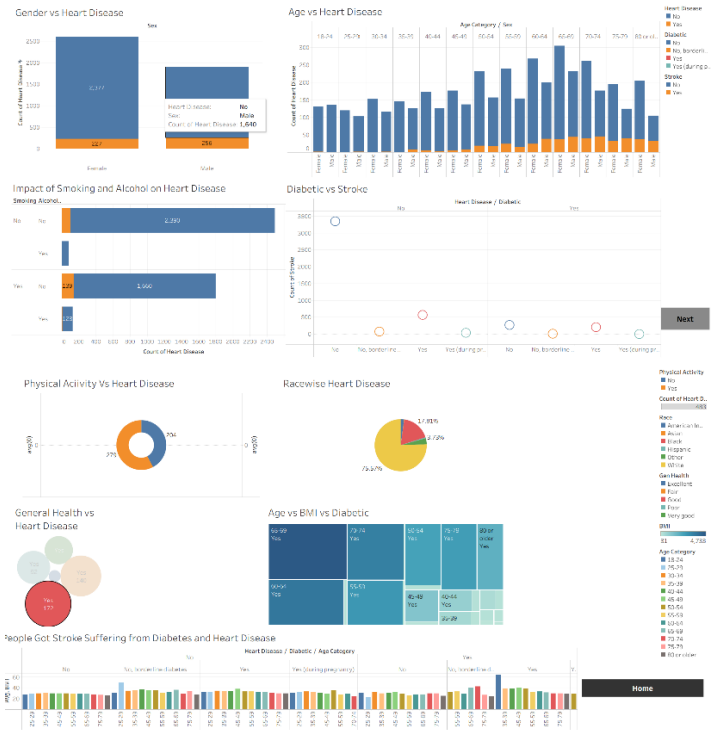
The performance testing phase evaluates how efficiently the Tableau dashboard processes, renders, and visualizes the dataset. The objective is to ensure that the dashboard provides fast, accurate, and interactive insights when users explore heart disease data.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	**Description:** The heart disease dataset was successfully imported into Tableau from a CSV file. The dataset contains patient health indicators such as BMI, smoking habits, physical activity, diabetes status, age category, and other medical conditions. Tableau processed the dataset and generated visualizations efficiently. **Observations:** • Dataset Size: ~4500 records • Number of Attributes: 18 columns • Data Types: Dimensions and Measures • Rendering Time: Less than 2 seconds for visualization loading **Result:** The data was rendered smoothly in Tableau worksheets and dashboards without performance issues. Visualizations update dynamically based on filters and selections. **Screenshot:** 

		
2.	Data Preprocessing	**Description:** Data preprocessing was performed to ensure the dataset was clean and suitable for visualization. Tableau Prep Builder was used to structure and validate the data before loading it into Tableau Desktop. **Steps Performed:** • Removed duplicate records • Checked for missing or null values • Converted categorical fields into dimensions • Validated numerical measures such as BMI and SleepTime • Standardized column names for better readability **Result:** The cleaned dataset improved dashboard performance and ensured accurate visualizations.
3.	Utilization of Filters	**Description:** Interactive filters were added to the dashboard to allow users to analyze heart disease data based on different health factors. **Filters Implemented:** • Age Category • Gender • Smoking Habit • Physical Activity • Diabetes Status • General Health Condition **Result:** Users can dynamically explore the dataset and identify patterns affecting heart disease risk. Filters update visualizations instantly without delay.

		Screenshot: 

		
4.	Calculation fields Used	**Description:** Calculated fields were created to derive meaningful insights from the dataset and categorize patient health conditions. **Calculated Fields Created:** **1. BMI Category** Example Calculation: IF [BMI] < 18.5 THEN "Underweight" ELSEIF [BMI] >= 18.5 AND [BMI] < 25 THEN "Normal" ELSEIF [BMI] >= 25 AND [BMI] < 30 THEN "Overweight" ELSE "Obese" END **2. Health Risk Indicator** IF [HeartDisease] = "Yes" THEN "High Risk" ELSE "Low Risk" END

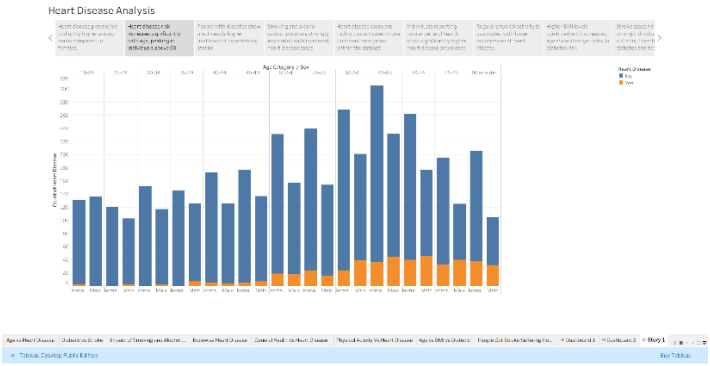
		Result: Calculated fields helped classify patients into different risk groups and improved the interpretability of visualizations.
5.	Dashboard design	**Description:** An interactive Tableau dashboard was designed to visualize patterns and trends related to heart disease. The dashboard integrates multiple charts to provide a comprehensive overview of patient health indicators. **Number of Visualizations / Graphs:** 6 **Visualizations Included:** 1. Heart Disease Distribution (Pie Chart) 2. Heart Disease by Age Category (Bar Chart) 3. Smoking vs Heart Disease (Stacked Bar Chart) 4. BMI Distribution (Histogram) 5. Physical Activity vs Heart Disease Risk (Bar Chart) 6. General Health Condition Analysis (Heatmap) **Result:** The dashboard provides a clear and interactive interface for analysing factors associated with heart disease risk. **Screenshot:** 
6	Story Design	**Description:** A Tableau Story was created to present analytical insights in a step-by-step format. This helps stakeholders understand patterns and key findings from the dataset. **Number of Visualizations / Graphs:** 5 **Story Flow:**

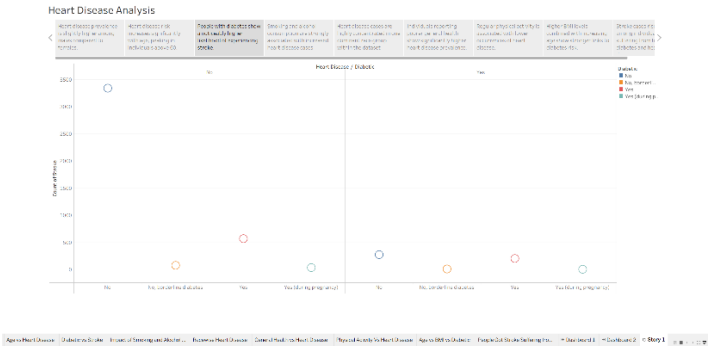
1. Introduction to Heart Disease Dataset
2. Demographic Analysis of Patients
3. Lifestyle Factors Affecting Heart Disease
4. Health Indicators and Risk Levels
5. Key Insights and Observations

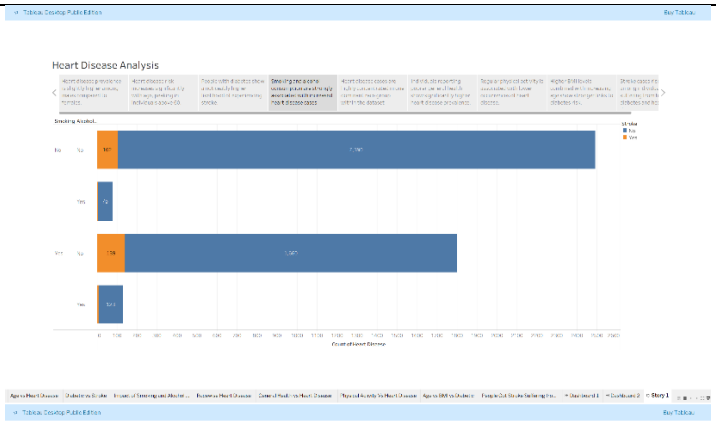
Result:

The Tableau story effectively communicates analytical insights and supports decision-making through visual storytelling.

Screenshot:



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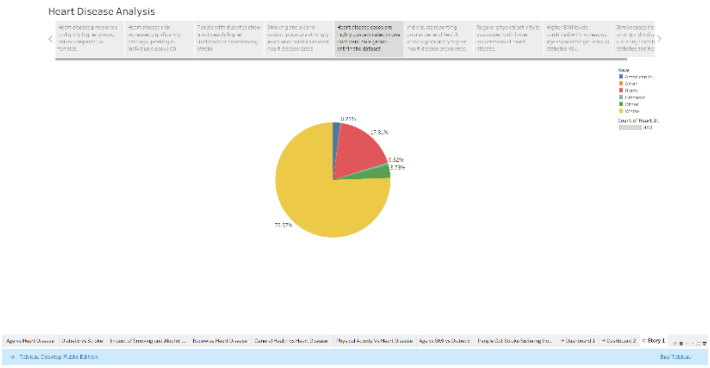
Count of Heart Dis...
Story 1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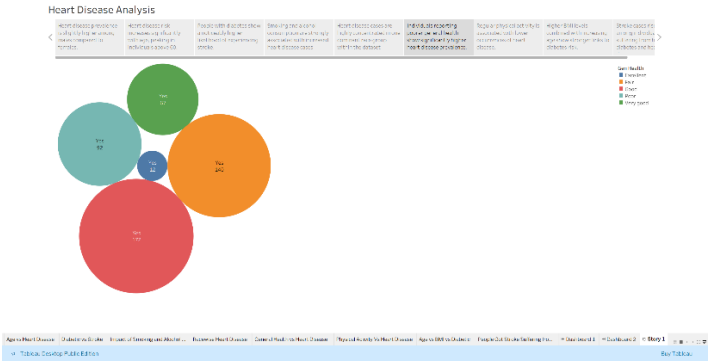
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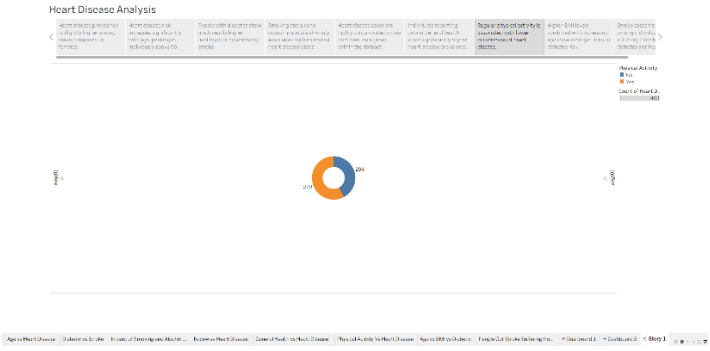
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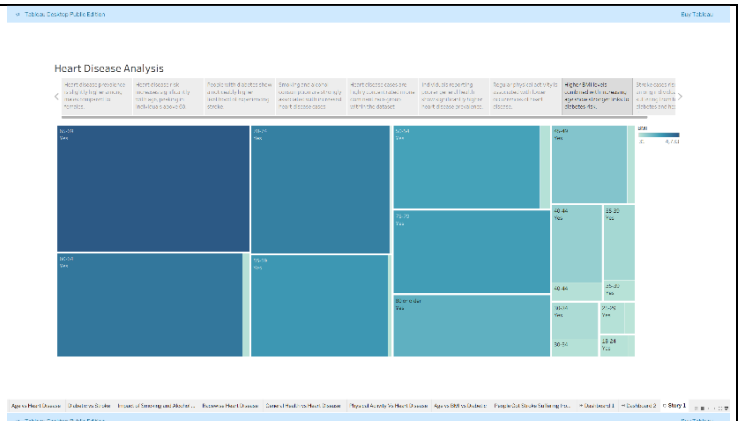
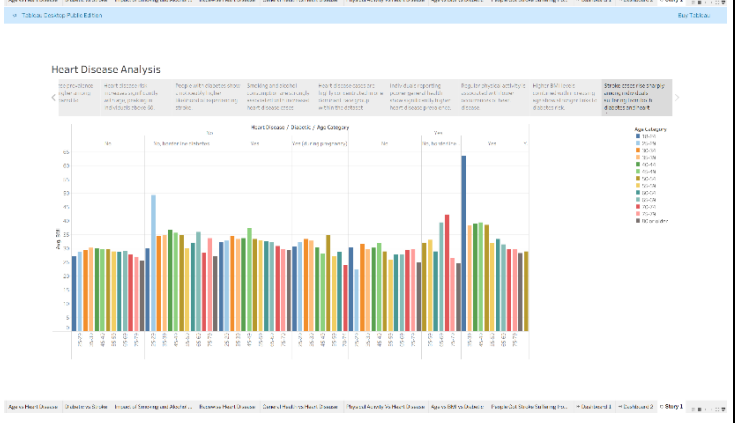
Story 1

		 Tableau Desktop Public Edition Buy Tableau Heart Disease Analysis Story 1  Tableau Desktop Public Edition Buy Tableau Heart Disease Analysis Heart Disease / Diabetic / Age Category Story 1